

Finding Equilibria in Farsighted Coalition Formation: A Computational Search Problem

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9 June 2026

Context

This work concerns a dynamic game-theoretic model of international coalition formation, developed by Heyen & Lehtomaa (2021) in the context of *solar geoengineering governance* [1]. Solar geoengineering refers to deliberate, large-scale interventions to reduce incoming solar radiation and thereby cool the planet. Countries differ in their preferred level of cooling, so any deployment level will benefit some and harm others.

We focus exclusively on the *power-threshold* governance regime, in which a coalition can only deploy geoengineering if its combined power (e.g. economic weight) meets a minimum threshold—say, 50% of global power. In practice this reflects the threat of international sanctions: a single country that attempts unilateral deployment faces sufficient economic or political pressure from the rest of the world to be deterred. Under this regime, no individual country can act as a “free driver”; geoengineering deployment requires at least a two-country coalition. The central question then becomes which *coalition structures*—agreements about who cooperates with whom—form and persist over time.

The model studies how governance rules—who can propose changes to the coalition structure, who must consent, and what transitions are feasible—affect which structures emerge. Crucially, countries are *farsighted*: when deciding whether to join or leave a coalition, they do not just evaluate the immediate payoff change, but anticipate the full chain of future state transitions that their decision sets in motion.

1 A Three-Player, Five-State Game

The canonical illustrative case features three countries—W (Warm), T (Temperate), and C (Cold)—with heterogeneous temperature preferences. Assuming equal power shares ($\frac{1}{3}$ each) and a 50% deployment threshold, at least two countries must cooperate to deploy geoengineering. The possible configurations of the international treaty system are captured by five *coalition structures* (states):

- $()$: all countries are singletons—no cooperation, no deployment ($G = 0$).
- (TC) : T and C cooperate and jointly deploy at their preferred level; W is a singleton.
- (WC) : W and C cooperate; T is a singleton.
- (WT) : W and T cooperate; C is a singleton.
- (WTC) : all three countries cooperate (grand coalition).

Each state delivers a fixed *static payoff* $u_i(x)$ to every player i , determined by which coalition controls geoengineering deployment and what level it chooses. Singleton countries outside the deploying coalition have no control over the outcome and simply receive whatever payoff the deployed level implies for them.

1.1 The Transition Graph as a Markov Chain

At each period, one country is selected as proposer (with equal probability $\frac{1}{3}$ each). The proposer may suggest a transition to a different state; a fixed *approval committee* then votes on whether to accept. Together, these proposal and acceptance decisions determine transition probabilities between states. Given a strategy profile σ , the resulting transition probability matrix P makes the sequence of states a *Markov chain* on the five-state space.

Figure 1 shows an example transition graph for a specific equilibrium strategy profile: self-loops indicate the probability of remaining in the current state, and directed edges give the probability of moving to another state in one period.

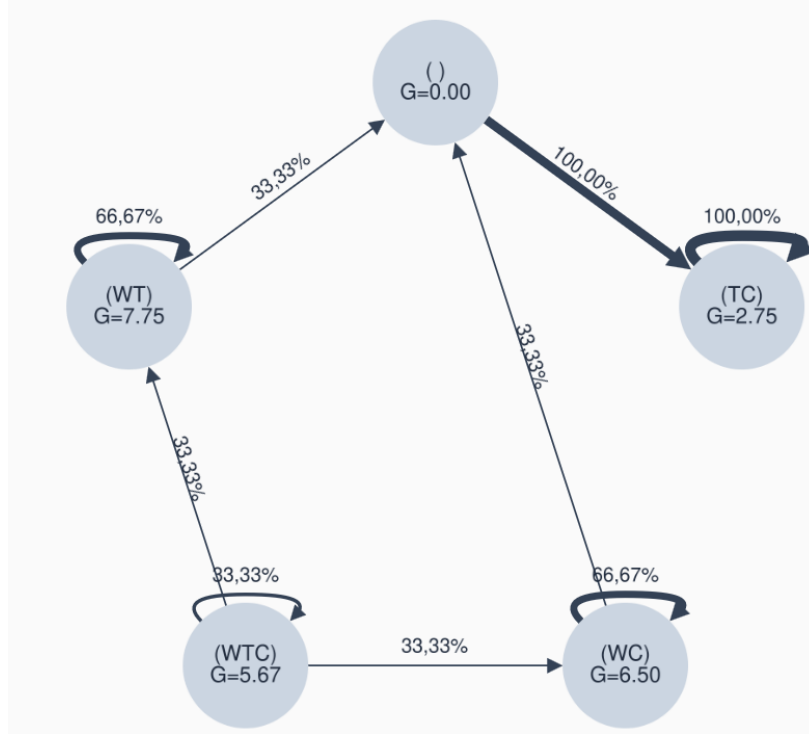


Figure 1: Equilibrium transition graph for the power-threshold scenario. Node labels show the state name and the long-run expected payoff G (geoengineering level). Edge weights are transition probabilities. States (TC) and (WTC) are absorbing under this equilibrium.

1.2 Strategy Profiles

A *strategy profile* σ encodes two types of decisions for every player i , state x , and possible next state y :

1. **Proposal probability** $\sigma_i^P(x, y) \in [0, 1]$: the probability that player i , when selected as proposer in state x , proposes a transition to state y . These must sum to one across all y for each (i, x) .

2. **Acceptance probability** $\sigma_i^A(x, y \mid j) \in [0, 1]$: when proposer j proposes to move from x to y , the probability that approval-committee member i votes in favour. Only committee members have a decision here; the remaining entries are undefined.

Table 2 shows the structure of a strategy profile as used in the solver. Rows correspond to (state, role, player) triples; columns correspond to (proposer, proposed next state) pairs. Orange cells mark the specific decisions taken in this example; grey cells indicate entries where the player is not in the approval committee.

			Proposer W					Proposer T					Proposer C				
			()	(TC)	(WC)	(WT)	(WTC)	()	(TC)	(WC)	(WT)	(WTC)	()	(TC)	(WC)	(WT)	(WTC)
()	Proposition	NaN	1					1					1				
()	Acceptance	W	1							1	1	1			1	1	1
()	Acceptance	T		1		0	0	1						1		0	0
()	Acceptance	C		1	0		0		1	0		0	1				
(TC)	Proposition	NaN	1					1					1				
(TC)	Acceptance	W		1						1	1	1			1	1	1
(TC)	Acceptance	T	0		0	0	0	0	1						0	0	0
(TC)	Acceptance	C	0		0	0	0			0	0	0	0	1			
(WC)	Proposition	NaN	1					1					1				
(WC)	Acceptance	W	0		1			0	0		1	1		0		1	1
(WC)	Acceptance	T		1		0	0			1				1		0	0
(WC)	Acceptance	C		1		0	0	1	1		0	0	1		1		
(WT)	Proposition	NaN	1					1					1				
(WT)	Acceptance	W	0			1			0	0		1	0	0	0		1
(WT)	Acceptance	T		1	1		1	1			1		1	1	1		1
(WT)	Acceptance	C		1	1		0		1	1		0				1	
(WTC)	Proposition	NaN	1					1					1				
(WTC)	Acceptance	W		0			1	0	0		0		0	0	0		
(WTC)	Acceptance	T	1		1	0				1		1	1	1	1		
(WTC)	Acceptance	C	1		1	1		1	1		1					1	1

Figure 2: Example strategy profile for the three-player game. Each block of rows covers one current state. Within each block, the first row is the proposer’s probability distribution over next states (“Proposition”), and the remaining rows are acceptance probabilities for each committee member (“Acceptance”). Grey/blank cells denote entries outside the approval committee. This example uses *pure strategies*: all entries are 0 or 1. In general, equilibria may involve *mixed strategies* with probabilities strictly between 0 and 1.

The rules governing *who* is in the approval committee for each transition are fixed by the governance structure (effectivity correspondence) and are not decision variables. For example, under unanimity rules, all current coalition members must consent to a merger; a country can always exit unilaterally. These rules are held constant throughout the search for equilibria.

2 The Equilibrium Conditions

The goal is to find a strategy profile σ that constitutes a *Markov Perfect Equilibrium* (MPE). This is not an optimisation problem in the usual sense: there is no single objective to maximise. Instead, two rationality conditions must hold *exactly simultaneously*.

Intuitively: **players must behave rationally with respect to their long-run payoffs.** A proposer should only suggest a transition if—taking into account the chance the committee will approve it—the move is expected to improve their continuation value. A committee member should approve a transition if and only if the destination state gives them a higher continuation value than the current one. The catch is that continuation values themselves depend on the strategies: changing who proposes what changes the Markov chain, which changes the long-run values, which changes what is rational to propose and approve. An equilibrium is a strategy profile where these two sides are mutually consistent.

Value functions. Given a strategy profile σ and its induced transition matrix P , the *Bellman continuation value* (long-run expected payoff) for player i in state x is the unique solution to

$$V_i(x) = (1 - \delta) u_i(x) + \delta \sum_y P(x, y) V_i(y), \quad (1)$$

where $\delta \in (0, 1)$ is a discount factor (typically $\delta = 0.99$). In matrix form, $V_i = (I - \delta P)^{-1} (1 - \delta) u_i$.

Condition 1 (Proposal rationality). A proposer only puts positive probability on transitions that maximise their expected continuation value (accounting for committee approval):

$$\sigma_i^P(x, y) > 0 \implies y \in \arg \max_{y'} [\sigma^A(x, y' | i) V_i(y') + (1 - \sigma^A(x, y' | i)) V_i(x)], \quad (2)$$

where $\sigma^A(x, y | i)$ is the probability that the committee approves the proposal.

Condition 2 (Approval rationality). A committee member approves if and only if the destination state improves their continuation value:

$$\sigma_k^A(x, y | i) = \begin{cases} 1 & \text{if } V_k(y) > V_k(x), \\ 0 & \text{if } V_k(y) < V_k(x), \\ \in [0, 1] & \text{if } V_k(y) = V_k(x). \end{cases} \quad (3)$$

A strategy profile is an equilibrium if and only if equations (1)–(3) hold simultaneously. There is no objective function to gradient-descend; the problem is to find a *fixed point* of this coupled system.

3 Size of the Strategy Space

To appreciate the difficulty, consider the number of distinct binary strategy profiles (setting each probability to 0 or 1):

- **Proposals:** For each of the $n = 3$ players and each of the $S = 5$ states, the player proposes one of $S = 5$ possible next states. This gives $5^{3 \times 5} = 5^{15} \approx 3 \times 10^{10}$ possible proposal rules. Expressed as binary indicator variables (one bit per (i, x, y) triple with the constraint that exactly one y is chosen), there are $3 \times 5 \times 5 = 75$ binary entries.
- **Acceptances:** Similarly, for each committee member, each proposed transition (i, x, y) requires a binary accept/reject decision. The number of such entries is comparable to 75, summing over all relevant (proposer, current state, next state, approver) tuples.

Treating all decisions as independent binary variables gives a strategy space of size approximately $2^{75} \times 2^{75} = 2^{150} \approx 10^{45}$. This is orders of magnitude larger than anything approachable by brute force.

4 Searching in the Ordinal Value Space

Rather than searching over raw strategies, a key structural insight allows a more tractable formulation. Under the equilibrium conditions, a player’s decisions depend only on the *ordinal ranking* of continuation values $V_i(x)$ across states x : an approval committee member votes yes if and only if the proposed state ranks higher, and a proposer proposes the highest-ranked state (conditional on approval probabilities). This motivates searching over *ordinal rankings of states* per player.

Figure 3 shows the continuation values $V_i(x)$ for each player i and state x in an example equilibrium. The equilibrium strategies are fully determined by the ordinal structure of these values: for instance, W ranks $(CTW) \succ (CW) \succ (CT) \succ () \succ (TW)$, so W approves any transition towards (CTW) and proposes it whenever selected. The search problem reduces to finding which such ordinal ranking triple is self-consistent—i.e. the values induced by the corresponding strategy profile reproduce the same ranking.

Long-term Value Functions (V) and Geoengineering Levels					
State	W	T	C	G (°C cooling)	Deployed by
()	53,460000	13,860000	0,660000	0,000000	None
(CT)	55,687500	14,437500	0,687500	2,750000	(TC)
(CW)	59,436667	13,403333	-1,941111	6,500000	(WC)
(TW)	62,632251	13,077954	-3,440144	7,750000	(WT)
(CTW)	63,746173	13,383210	-3,404444	5,666667	(WTC)

Figure 3: Long-run continuation values $V_i(x)$ and geoengineering deployment levels for each state x and player $i \in \{W, T, C\}$ in an example equilibrium. The equilibrium conditions depend only on the *ordinal* ranking of these values within each row (per player): which state a player prefers to be in determines both approval and proposal behaviour. Highlighted cells mark values that are decisive for a player’s ranking.

Strict ordinal rankings. For $n = 3$ players and $S = 5$ states, there are $5! = 120$ strict rankings (permutations) per player, giving a search space of

$$(5!)^3 = 120^3 = 1,728,000 \approx 1.7\text{M}$$

triples of rankings to examine. For each triple, one can: (1) induce a strategy profile deterministically, (2) compute V via the linear system (1), and (3) verify whether conditions (2)–(3) hold. This is exhaustively feasible.

Weak ordinal rankings (allowing ties). Since value ties are possible (two states may deliver the same continuation value to a player), one must also consider *weak orders*, where ties between states are permitted. The number of weak orders on $S = 5$ states is given by the ordered Bell (Fubini) number $a(5) = 541$. The resulting search space is

$$541^3 = 158,340,421 \approx 150\text{M}.$$

Computational cost. A full exhaustive search over the $\approx 150\text{M}$ weak-order triples takes up to **24 hours** on a 2-CPU virtual machine. For larger games—e.g. $n = 4$ players, $S = 15$ states, or $n = 5$ players, $S = 52$ states—the search space grows as $(a(S))^n$ and quickly becomes intractable.

Density of equilibria. In practice, for the three-player scenarios studied, the search typically finds **1–10 equilibria** among the $\approx 150\text{M}$ candidates. The density is thus on the order of 10^{-7} , making random sampling highly unlikely to succeed.

5 The Core Algorithmic Challenge

The exhaustive ordinal-ranking search solves the problem for $n = 3$, but does not scale. The natural alternative is an iterative fixed-point algorithm:

1. Start from an initial strategy profile $\sigma^{(0)}$ (e.g. random or uniform).
2. Compute continuation values $V^{(t)}$ from $\sigma^{(t)}$ via (1).
3. Update the strategy profile $\sigma^{(t+1)}$ to be rational given $V^{(t)}$ (i.e. apply conditions (2)–(3)).
4. Repeat until convergence.

The difficulty is that this iteration has no guaranteed convergence properties. In practice, it frequently *cycles*: the strategy update at step 3 changes P , which changes V , which changes the rational strategy, which may return to a previously visited profile without ever reaching a fixed point. Smoothed/softmax approximations (replacing hard argmax and step functions with temperature-parameterised soft versions, followed by annealing) partially mitigate cycling but are unreliable, particularly for larger n or scenarios with a sparse equilibrium structure.

Open question. We are looking for a principled algorithm—possibly iterative, possibly based on branch-and-bound, mixed-integer programming, or other techniques from reinforcement learning or game-theory computation—that can:

- Certifiably find *one* or *all* equilibria for a given game instance,
- Scale to at least $n = 4$ –5 players, and
- Handle the fixed-point structure of the problem without cycling.

The problem has similarities to finding Nash equilibria in stochastic games, value iteration in MDPs with endogenous transition kernels, and feasibility problems in complementarity programming—but the specific structure of endogenous Markov transitions, coupled proposal and approval decisions, and the ordinal rationality conditions makes it non-standard.

References

- [1] D. Heyen and J. Lehtomaa, “Solar geoengineering governance: A dynamic framework of far-sighted coalition formation,” *Oxford Open Climate Change*, vol. 1, no. 1, 2021.